

Authored Grade-9 Force and Laws of Motion spine covering balanced/unbalanced for: Construction

Zero net force allows steady motion

You need

- Can tell whether an object's velocity is changing.

The key step

Zero net force does not mean zero motion; it means velocity does not change, so rest and steady motion are both compatible with the same condition.

Separate 'is moving' from 'is changing its velocity', and link zero net force only to the second.

1. **Separate the statement 'is moving' from the statement 'is changing its velocity.'**

Why valid: Motion and change of motion are different observations, and only the second identifies acceleration.

Result: state A: moving steadily. state B: speeding up, slowing down, or turning.

2. **Imagine a low-friction cart after the hand has stopped touching it and compare successive equal-time snapshots.**

Why valid: Once the hand is gone, a continuing steady velocity can be observed without a continuing

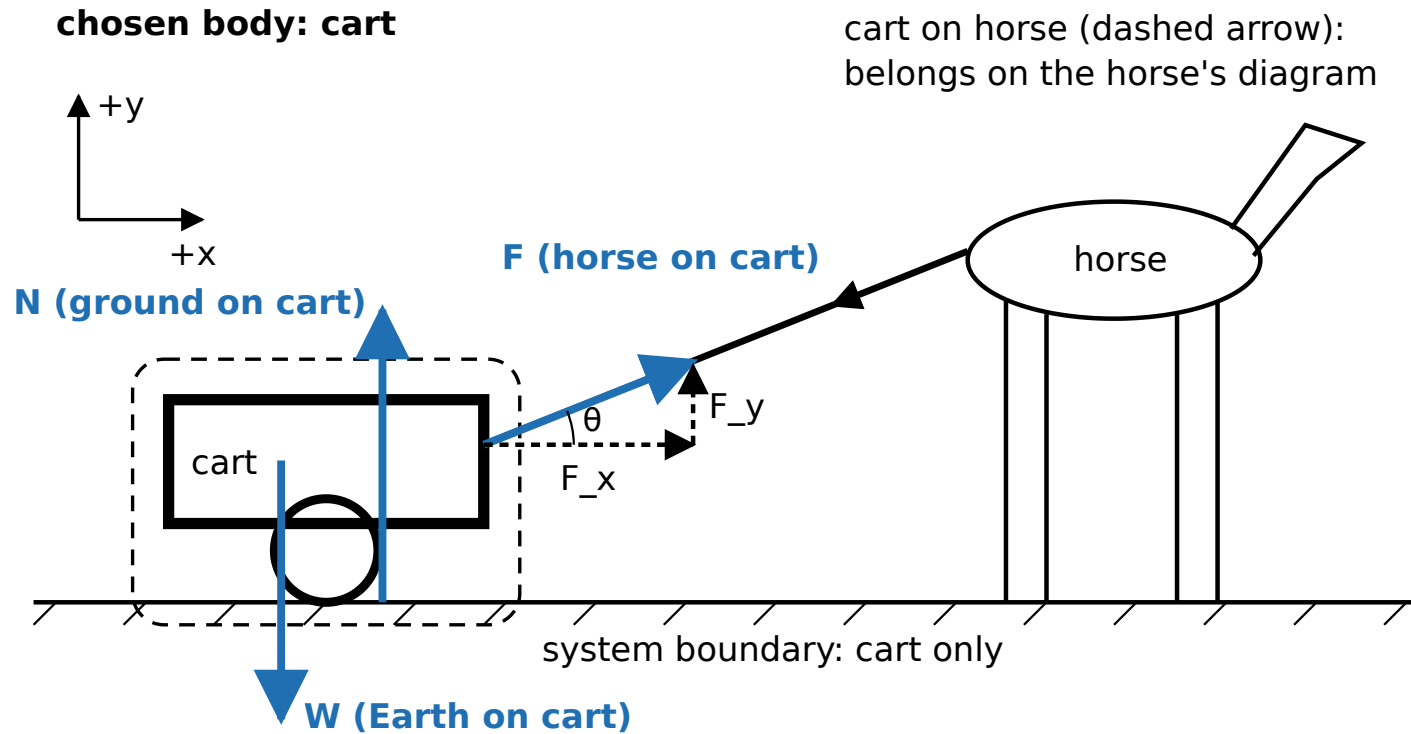
forward push.

Result: successive equal-time snapshots -> equal forward position advances; velocity unchanged.

3. **Compare the released moving cart with a cart that starts at rest under the same zero-net-force condition.**

Why valid: If both keep their initial velocities, the rule is about unchanged velocity rather than about rest.

Result: moving cart: stays steady. resting cart: stays at rest. both show no velocity change.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

An ice-hockey puck of mass 0.2 kg slides across smooth ice at 4 m/s after the stick loses contact. Friction and air resistance are negligible. What horizontal forces act on it, what is its velocity 3 s later, and would a puck at rest on the same ice start moving?

1. Once the stick loses contact, nothing touches the puck horizontally, so the horizontal net force is zero.
2. $F_{\text{net}} = m a$ gives $a = 0/0.2 = 0 \text{ m/s}^2$, so the velocity does not change: after 3 s it is still 4 m/s.
3. Equal times give equal advances: 4 m in each second, 12 m in 3 s.
4. A puck at rest under the same zero net force keeps its zero velocity, so it stays at rest.

No horizontal force acts. The puck still moves at 4 m/s, having covered 12 m, and a puck at rest stays at rest.

A tempting wrong path

- A force must keep acting in the direction of motion or the object must stop.

Diagnose

- A low-friction cart is pushed and then released. After the hand loses contact, what horizontal force is required merely to keep it moving steadily?

Repair

- Separate velocity from change of velocity: a net force is required for acceleration, not for the existence of motion.

Check it independently

- Put back a small friction force: the object must now slow down, which shows that a force changes the velocity rather than maintaining it.
- $a = F_{\text{net}}/m$ gives $0 \text{ N} / m = 0 \text{ m/s}^2$, and distance equals speed $\times$ time in metres for equal time intervals.

Exit task

One cart is at rest and another moves steadily in a straight line. For each, decide whether zero net force is compatible with the motion and explain the common rule.

Model answer

Zero net force is compatible with both. Each cart can keep its current velocity unchanged.

1. Rest is a zero velocity that stays zero.
2. Steady straight motion is a nonzero velocity that stays unchanged.
3. The shared condition is no change of velocity, not no motion.

Nonzero forces can cancel

You need

- Can interpret zero net force as unchanged velocity.
- [Relate zero net force to unchanged velocity.](#)

The key step

Several nonzero forces can act on one body while their directed sum is zero; the motion responds to the net result rather than to the mere presence of forces.

List every force on the body as its own contribution, then combine them with signs; a zero sum never means that no forces act.

1. **List two opposite pushes on the same object as separate contributions before combining them.**

Why valid: Keeping the contributions visible prevents a zero total from being mistaken for no forces.

Result: leftward push: present. rightward push: present.

2. Make the two opposite pushes equal in size and combine their directional effects.

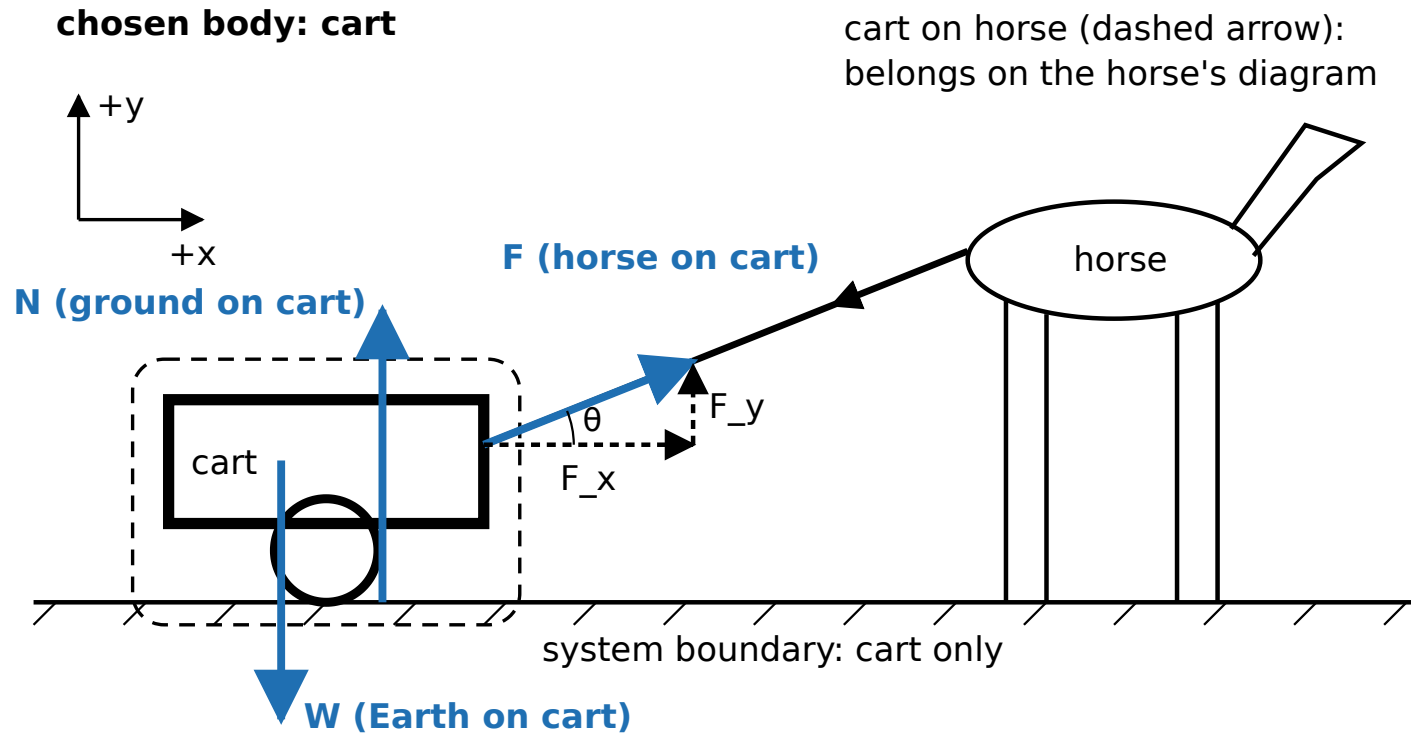
Why valid: Equal opposite contributions cancel in the net result while each contribution still exists.

Result: two nonzero opposite pushes $\rightarrow$ net force: zero.

3. Increase both opposite pushes together while keeping them equal.

Why valid: The individual forces change while the zero net result persists, directly separating the parts from their sum.

Result: larger opposite pushes: still net force zero; individual forces remain nonzero.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

A 1.5 kg lamp hangs at rest from a ceiling cord. Take $g = 10 \text{ m/s}^2$ and up as positive. List each force on the lamp with its size, find the net force, and say what would happen if the cord were cut.

1. Two agents act on the lamp: the Earth, through its weight, and the cord, through its tension.
2. Weight: $1.5 \times 10 = 15 \text{ N}$, so -15 N on the declared axis.
3. The lamp is at rest, so $F_{\text{net}} = 0$: $T - 15 = 0$ and $T = +15 \text{ N}$. Both forces are real and non-zero.
4. Cut the cord and only the weight remains: $F_{\text{net}} = -15 \text{ N}$ and $a = -15/1.5 = -10 \text{ m/s}^2$.

The cord pulls up with 15 N and gravity pulls down with 15 N; the net force is zero. If the cord were cut, the lamp would accelerate downward at 10 m/s^2 .

A tempting wrong path

- If the net force is zero, no forces are acting.

Diagnose

- A book rests on a table. Are there no forces on the book, or can nonzero forces add to zero?

Repair

- List every interaction force on the body, then combine their directions; cancellation belongs to the net result, not to the individual forces.

Check it independently

- Remove one of the balancing forces: the other must now produce an acceleration, which proves it was acting all along.
- With one axis declared, the balancing forces must carry opposite signs and equal sizes; two forces of the same sign can never sum to zero.

Exit task

An object has two equal opposite forces acting on it. Explain what is zero and what is not, then state what motion change the net result predicts.

Model answer

The net force is zero, but both individual forces are nonzero. The zero net result predicts no change of velocity.

1. Each force remains an actual interaction on the body.

2. Opposite equal directional effects cancel only when combined.
3. A zero net force is consistent with unchanged velocity.

A free-body diagram belongs to one chosen body

You need

- Can keep several forces visible even when their net effect is zero.

The key step

A free-body diagram contains forces acting on one chosen body only; the equal-and-opposite partner of an interaction acts on the other body and belongs on that other body's diagram.

Pick one body and admit only forces that answer 'who acts on this body?'; a partner force, or the components of a force already listed, never joins that list as an extra force.

1. **Choose the cart as the body to describe and require every arrow to answer 'who acts on the cart?'**

Why valid: The ownership question gives one mechanical test for whether a force belongs on the chosen body's diagram.

Result: chosen body: cart. every listed arrow must be an agent → cart force.

2. **For the horse-cart contact, place the horse-on-cart force on the cart diagram and place the cart-on-horse partner on a separate horse diagram.**

Why valid: The two forces belong to one interaction but act on different bodies, so combining them on one body's diagram changes the question being asked.

Result: cart diagram: horse $\rightarrow$ cart. horse diagram: cart $\rightarrow$ horse.

3. **Switch the chosen body from cart to horse and rebuild the inventory from the ownership question.**

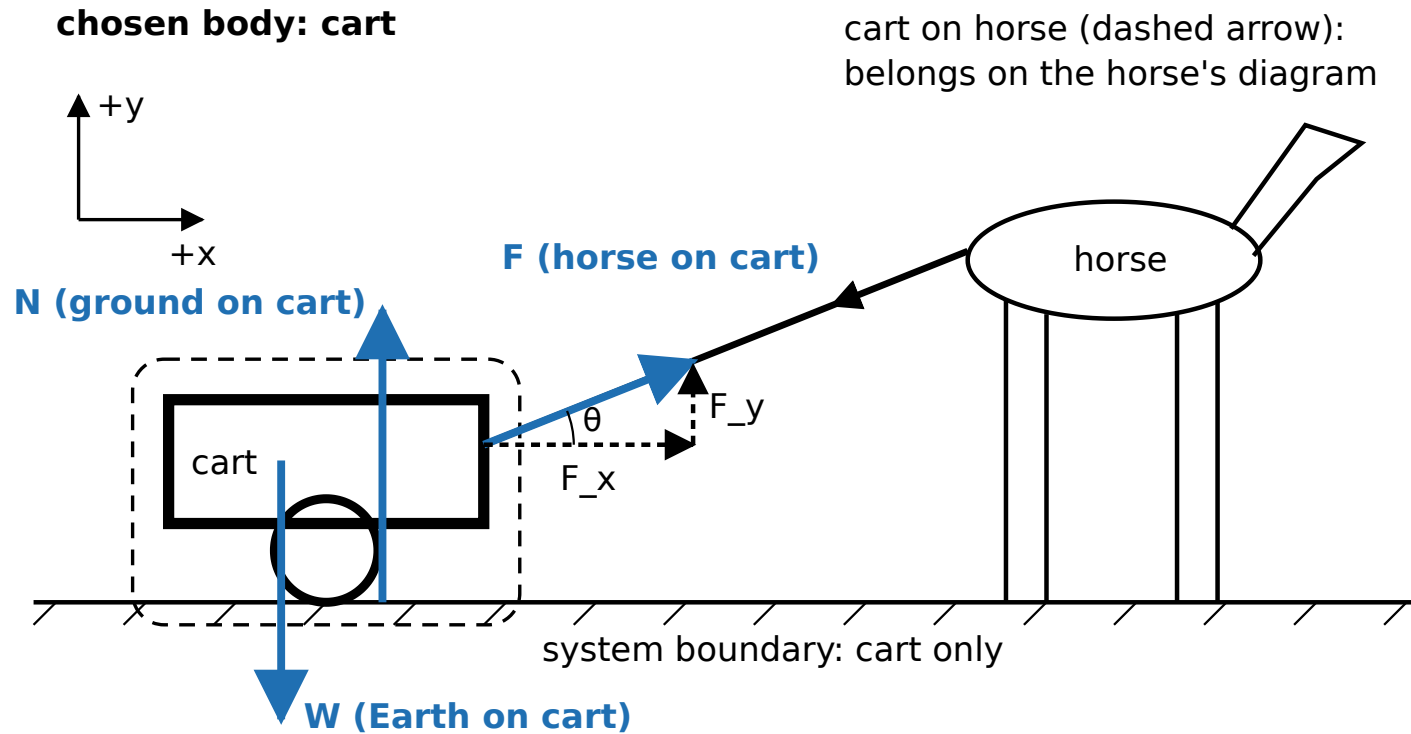
Why valid: A correct diagram changes when the chosen body changes even though the physical interaction is the same.

Result: switch body $\rightarrow$ switch which contact-force arrow belongs in the diagram; partner arrows never cancel on one diagram.

4. **When a physical force is resolved into components for chosen axes, keep the body-force inventory unchanged: use the components as a representation of that one force in the equations, not as additional interaction forces alongside the original vector.**

Why valid: Resolving a vector changes its coordinate representation, not the number of physical interactions acting on the body. Counting both the original force and its components double-counts the same interaction.

Result: one physical force $\rightarrow$ either original vector representation or its signed components in the chosen equations, never both as separate forces



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

A 50 kg child stands on a 20 kg box that rests on the floor. Take $g = 10 \text{ m/s}^2$. Build the free-body inventory of the box: name each force, its agent, its direction and its size.

1. Chosen body: the box. Every arrow must answer 'who acts on the box?'.
2. Earth on box: weight $20 \times 10 = 200 \text{ N}$ down. Child on box: a contact push down; the child is at rest, so it equals the child's weight, 500 N.
3. Floor on box: the box is at rest, so the floor pushes up with $200 + 500 = 700 \text{ N}$.
4. The child's 500 N weight acts on the child, and the box's 500 N upward push acts on the child too; both belong on the child's diagram.

On the box: its weight 200 N down (Earth), the child's push 500 N down (child), and the floor's push 700 N up (floor). The child's own weight is not on the box's diagram.

A tempting wrong path

- Equal and opposite partner forces cancel each other on one body's free-body diagram.
- After resolving weight on an incline, mg , $mg \sin(\theta)$ and $mg \cos(\theta)$ are three separate forces on the block.

Diagnose

- A horse pulls a cart and the cart pulls back equally. Which of those two forces acts on the cart, and which acts on the horse?
- If gravity is one Earth-on-block interaction, did choosing tilted axes create two new physical agents?

Repair

- Choose one body first. Put only forces acting on that body on its diagram; draw the partner force only when the other body is chosen.
- No. Weight remains one physical force. Its parallel/perpendicular components are the coordinate representation used in the chosen equations; do not add the original mg again to those component sums.

Check it independently

- Treat the touching bodies as one system: the external supporting force it predicts must equal the one found from the single body's own diagram.
- Each interaction contributes exactly one arrow to a given diagram; a diagram holding both members of a pair has one arrow that belongs to the other body.

Exit task

A person pushes a box. Describe the two contact-force partner arrows and state which one belongs on the box's free-body diagram.

Model answer

The person's force on the box belongs on the box diagram. The box's equal-and-opposite force on the person belongs on the person's diagram.

1. Choose the box first.
2. Ask which member of the contact pair acts on the box.
3. The partner acts on the person, so it cannot cancel the first force on the box diagram.

An accelerating observer needs a declared pseudo-force convention

You need

- Can build a force inventory for one chosen body.
- [Assign each force arrow to the body on which that force acts.](#)

The key step

A force-based account depends on the observer: a non-accelerating observer uses the physical interaction forces, while an accelerating observer must add the corresponding pseudo-force convention to describe the same observed motion consistently.

Say whether the observer is accelerating; keep the physical forces identical in both descriptions and add $-m a_{\text{frame}}$ only for the accelerating observer.

1. **Describe the same loose object first from the roadside and then from inside a bus, stating whether each observer is accelerating.**

Why valid: The observer choice is part of the model; leaving it implicit makes two different

descriptions look contradictory.

Result: roadside: observer acceleration zero. bus during start: observer acceleration forward.

2. **Keep the physical interactions unchanged and compare what the accelerating bus observer sees the loose object do relative to the bus.**

Why valid: The apparent backward motion comes from the observer's changing motion, not from a newly discovered physical contact on the object.

Result: bus accelerates forward -> loose object appears to shift backward relative to the bus.

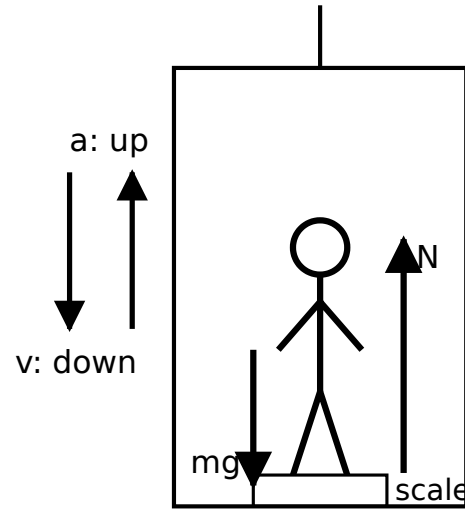
3. **Add a backward pseudo-force only in the accelerating-observer description and compare it with the roadside account.**

Why valid: The added convention compensates for the observer's acceleration without inventing a new physical interaction.

Result: roadside account: physical forces only. accelerating-bus account: same physical forces plus backward pseudo-force.

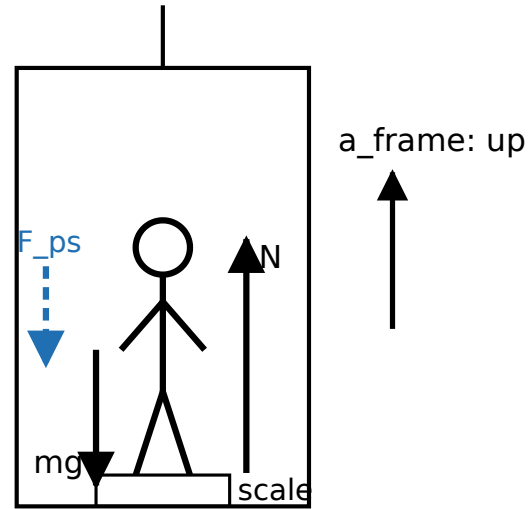
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Stage 1 of 5

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

Worked example

A 60 kg person stands on a scale in a lift that accelerates upward at 2 m/s^2 . Take $g = 10 \text{ m/s}^2$ and up as positive. Find the scale reading using the ground frame, then again using the lift frame with a pseudo-force, and compare.

1. Ground frame: the physical forces are the scale's push N up and the weight 600 N down, and the person accelerates at $+2 \text{ m/s}^2$.
2. Ground-frame equation: $N - 600 = 60 \times 2$, so $N = 720 \text{ N}$.
3. Lift frame: the person is at rest, and the frame accelerates at $+2 \text{ m/s}^2$, so add $F_{\text{pseudo}} = -ma_{\text{frame}} = -120 \text{ N}$.
4. Lift-frame equation: $N - 600 - 120 = 0$, so $N = 720 \text{ N}$, the same reading.

Both descriptions give a scale reading of 720 N .

A tempting wrong path

- The same ordinary-force diagram can be used unchanged in any accelerating observer's frame.

Diagnose

- A loose mass appears to accelerate backward inside a bus that starts forward. Can the bus observer use only the same physical-force inventory as the roadside observer?

Repair

- State the observer first. Use physical interactions alone for a non-accelerating observer; for an accelerating observer include the corresponding pseudo-force consistently.

Check it independently

- Both descriptions must predict the same observable, such as the scale reading or the string direction.
- Set the frame's acceleration to zero: the pseudo-force must vanish and the two descriptions must become identical.

Exit task

A bus accelerates forward and a loose object appears to move backward relative to the bus. Compare the roadside and bus-observer force descriptions without calculating any magnitude.

Model answer

The roadside description uses the physical interactions only. The accelerating bus description adds a backward pseudo-force convention to account for the observer's acceleration.

1. The physical interactions on the object do not change merely because the observer changes.
2. The bus observer is accelerating relative to the roadside observer.
3. The pseudo-force belongs to that accelerating-observer description rather than to a new physical agent.

Net force, not any one force, sets acceleration

You need

- Can choose one body, draw its external forces, and distinguish zero from nonzero net force.
- [Assign each force arrow to the body on which that force acts.](#)

The key step

For a chosen body of positive mass in the Grade 9 inertial-frame model, acceleration points with the signed net external force and satisfies $F_{\text{net}} = m a$.

Choose one body and a signed axis, add every external force on that body, and divide the net force by the mass; on a fixed incline, choose axes along and across the contact.

1. **Choose one body, declare a positive direction, and list only external forces acting on that body.**

Why valid: The net-force relation applies to one body's external force sum.

Result: one body; one signed axis; signed external-force list.

2. Add the signed forces to obtain F_{net} , then use $F_{\text{net}} = m a$.

Why valid: The net external force on the chosen body sets its acceleration: $F_{\text{net}} = m a$ holds for that one body, with $m > 0$, in an inertial frame.

Result: $F_{\text{net}} = \text{signed sum}$; $a = F_{\text{net}} / m$.

3. Return to an equal-opposite force case.

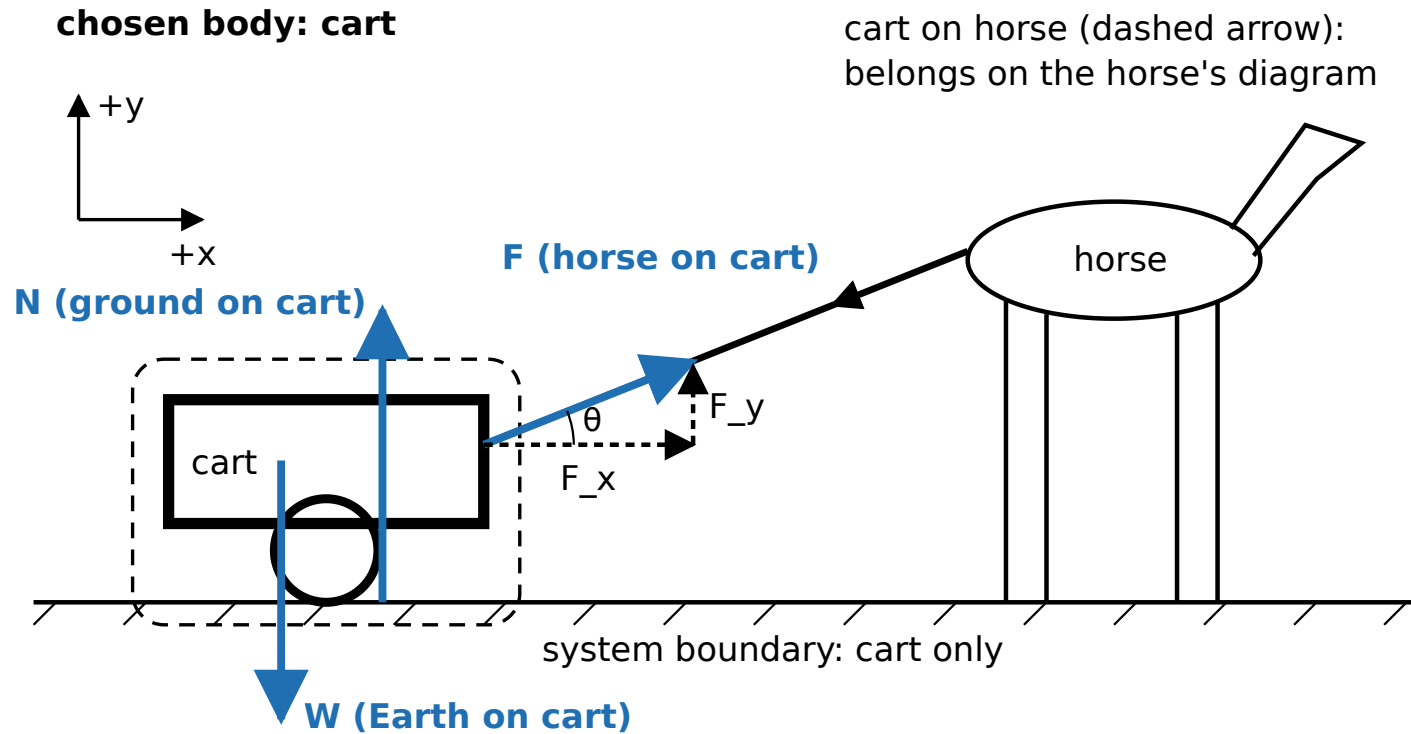
Why valid: The relation must recover zero acceleration when the signed force sum is zero.

Result: $F_{\text{net}} = 0 \rightarrow a = 0$ for $m > 0$.

4. Choose axes that make the active constraint easy to express; for a fixed inclined contact, use parallel/perpendicular axes when helpful, resolve the physical forces into those axes, and write separate signed Newton-II equations.

Why valid: Newton II is coordinate-independent. Aligning one axis with the contact makes maintained contact explicit as $a_{\text{perp}}=0$ while leaving the tangential acceleration free to be nonzero.

Result: contact-aligned model: $\Sigma F_{\text{parallel}} = m a_{\text{parallel}}$; $\Sigma F_{\text{perp}} = m a_{\text{perp}}$, with maintained fixed contact $\rightarrow a_{\text{perp}} = 0$



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

A 10 kg crate is pulled up a smooth ramp inclined at 30° by a rope parallel to the ramp with a force of 70 N. Take $g = 10 \text{ m/s}^2$. Using axes along and perpendicular to the ramp, find the crate's acceleration and the normal force, and the pull that would give zero acceleration.

1. Body: the crate. +x up the ramp, +y out of the ramp. Forces: the 70 N pull along +x, the 100 N weight vertically down, the normal force N along +y.
2. Resolve the weight: $100 \times \sin 30^\circ = 50 \text{ N}$ down the ramp and $100 \times \cos 30^\circ \approx 86.6 \text{ N}$ into the ramp.
3. Along the ramp: $F_{\text{net}} = 70 - 50 = 20 \text{ N}$, so $a = 20/10 = 2 \text{ m/s}^2$ up the ramp.
4. Across the ramp the crate keeps contact, so $a_y = 0$ and $N \approx 86.6 \text{ N}$. A 50 N pull would make $F_{\text{net}} = 0$ along the ramp, so $a = 0$.

$a = 2 \text{ m/s}^2$ up the ramp, $N \approx 86.6 \text{ N}$, and a 50 N pull would give zero acceleration.

A tempting wrong path

- Acceleration points with whichever individual force looks largest in the diagram.
- If a block on a fixed incline has zero acceleration perpendicular to the plane, its total acceleration must be zero.

Diagnose

- Two 10 N forces act opposite each other on a body. Does either force make the body accelerate in its direction?
- Can a block remain in contact with a fixed plane while speeding up down the plane?

Repair

- Add all signed external forces first; acceleration follows the net force, not an isolated arrow.
- Yes. Maintained contact sets only $a_{\text{perp}}=0$. The parallel component a_{parallel} is determined independently by the net force along the plane.

Check it independently

- The net force in newtons divided by the mass in kilograms must give m/s^2 .
- Flatten the incline to 0° : the method must give $N = mg$ and no down-slope weight component, the level-floor case.

Exit task

A 4 kg body has a net external force of +12 N along a declared axis. Find its acceleration.

Model answer

The acceleration is $+3 \text{ m/s}^2$.

1. Use $F_{\text{net}} = m a$ for the chosen body.
2. $a = +12 \text{ N} / 4 \text{ kg} = +3 \text{ m/s}^2$.
3. The positive sign matches the net-force direction.

Friction opposes relative sliding at a contact

You need

- Can choose one body and identify external contact interactions acting on it.
- [Assign each force arrow to the body on which that force acts.](#)

The key step

Friction is a contact force along the surface that opposes relative sliding, or the tendency for relative sliding, between the contacting surfaces; it is not defined by the object's overall velocity alone.

Before giving friction a direction, imagine the contact made perfectly slippery and ask which way the surfaces would slide relative to each other; friction opposes that, and is zero if there is no such tendency.

1. **Choose the body and mark the contact surface before deciding a friction direction.**

Why valid: Friction belongs to a specific contact interaction, so its direction must be read from relative motion at that contact.

Result: chosen body; named contact; tangent direction available.

2. **Imagine the relative sliding or tendency to slide if friction were absent, then point friction opposite that relative tendency.**

Why valid: Contact friction resists relative sliding between the surfaces.

Result: belt at 2 m/s, box at 0.5 m/s -> box slips left relative to the belt -> friction on the box points right

3. **Check a conveyor-belt case where the belt can frictionally accelerate an object in the same direction as the object's motion.**

Why valid: The counterexample falsifies the shortcut 'friction always points opposite velocity'.

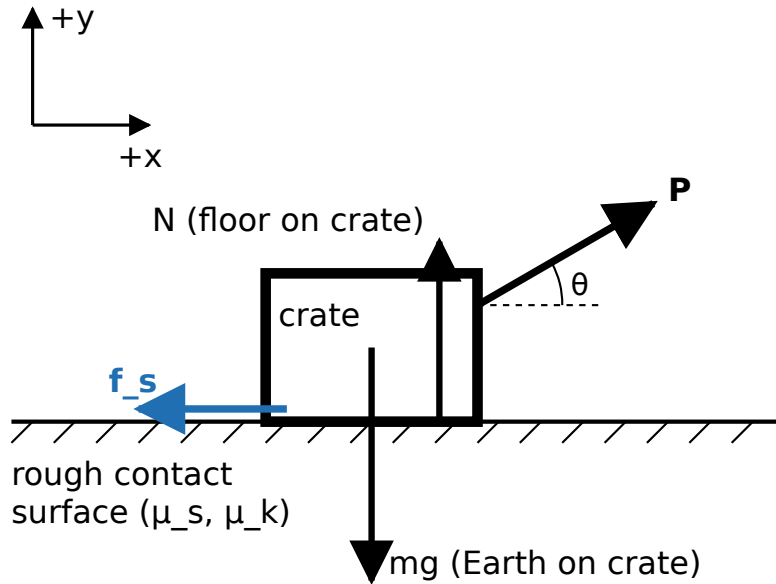
Result: friction direction follows contact slip tendency, not a universal page-direction rule.

4. **Decide whether any tangential friction is required at all; allow friction to be zero when the contact has neither relative sliding nor a tendency to slide.**

Why valid: Friction is a responsive tangential contact force, not an automatic nonzero force merely because two rough surfaces touch.

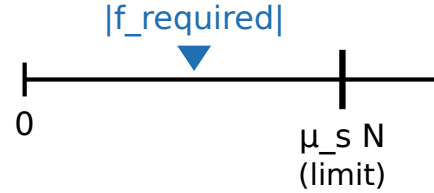
Result: contact-state check -> friction may be zero when no tangential force is required.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



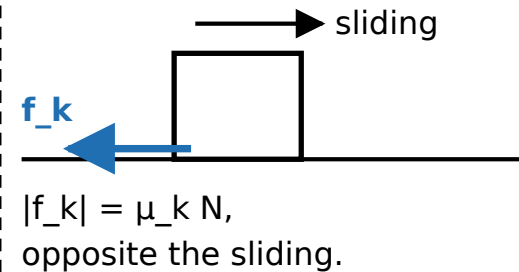
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Worked example

A book rests on a tray. A waiter first carries the tray at steady speed in a straight line, then speeds up forward; the book never slips. Find the friction force of the tray on the book in each stage, including whether it can be zero.

1. Choose the book as the body; the contact is the tray's top surface, and friction acts along it.
2. Steady carry: if the tray became perfectly slippery, the book would keep moving with the tray, so there is no slip tendency and the friction is zero.
3. Speeding up: without friction the book would keep its old speed while the tray moved ahead, so the book would slip backward relative to the tray.
4. Static friction opposes that tendency, so the tray's friction on the book points forward, the same way the book is moving.

During the steady carry the friction is zero; while the tray speeds up, static friction on the book points forward.

A tempting wrong path

- Friction always points opposite the object's velocity.
- Two rough surfaces in contact must always exert a nonzero friction force.

Diagnose

- A moving conveyor belt speeds up a box resting on it. Can the friction on the box point in the same direction as the box's motion?
- A crate and rough platform already move together at constant velocity. If no horizontal force is otherwise required, must the friction force be nonzero merely because the surfaces are rough?

Repair

- Decide the relative sliding tendency at the contact first; friction opposes that tendency.
- No. Ask what relative sliding or slip tendency would occur without friction. If none exists and no tangential force is required, friction can be zero.

Check it independently

- Compare with Newton's second law: if friction is the only horizontal force on the body, it must point the way the body's velocity is changing.
- Remove every sideways push and every change of speed: the friction you predict must fall to zero.

Exit task

A box is placed on a conveyor belt moving to the right faster than the box. Before the box matches the belt speed, what direction is the friction force on the box?

Model answer

The friction force on the box points to the right.

1. Relative to the belt, the box tends to slip left at the contact.
2. Friction opposes that relative slip tendency.
3. Therefore the belt's friction on the box points right and accelerates it.

Third-law partner forces act on different bodies

You need

- Can assign force arrows to the body on which they act.
- [Assign each force arrow to the body on which that force acts.](#)

The key step

One interaction produces two partner forces of equal magnitude and opposite direction, but they act on different bodies and therefore do not cancel in a single body's net-force sum.

Name each force of an interaction as 'A on B' or 'B on A', and put it on the diagram of the body it acts on, so the pair never meets in one sum.

1. **Name the interaction as 'A on B' and 'B on A' before drawing arrows.**

Why valid: The agent-target wording records which body owns each partner force.

Result: force of A on B belongs to B; force of B on A belongs to A.

2. **Draw equal-length opposite arrows on the two different body diagrams.**

Why valid: Newton's third-law partners are equal in magnitude and opposite in direction for the same

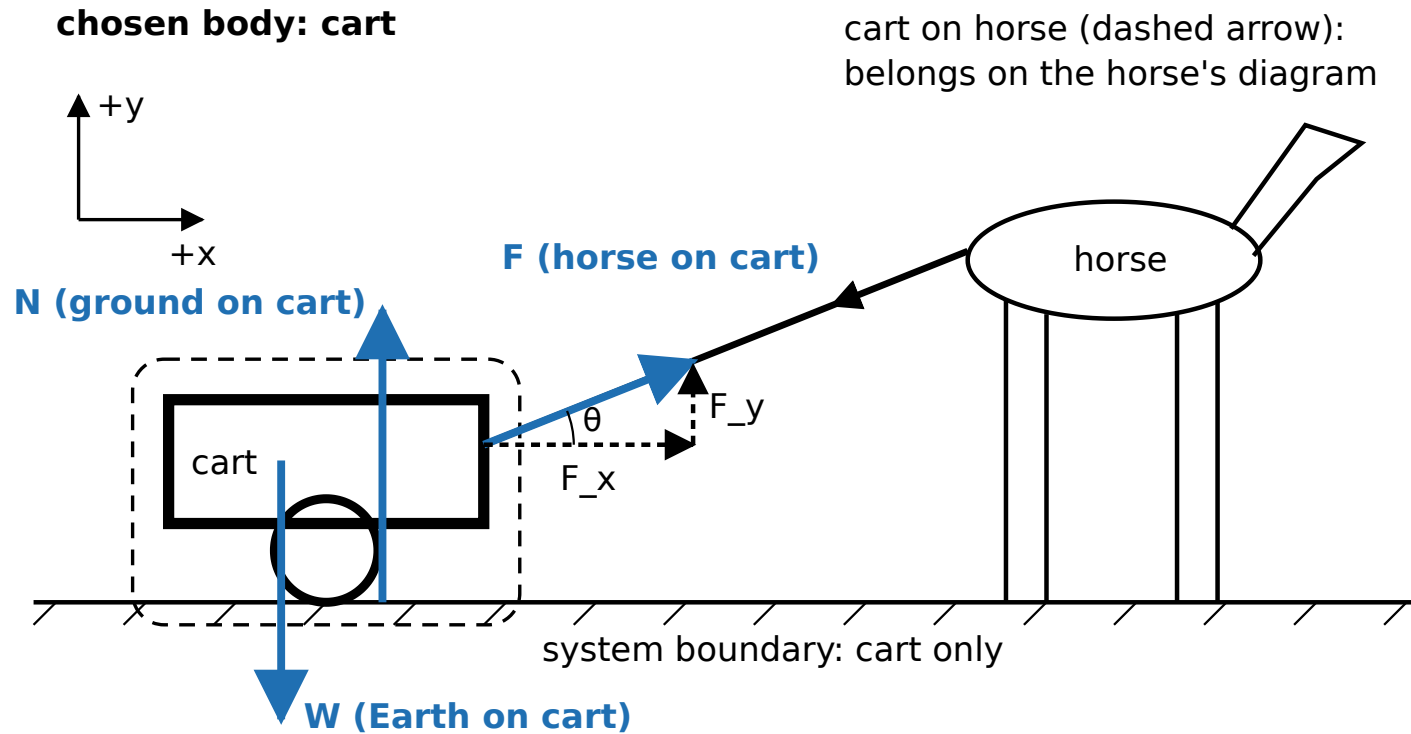
interaction.

Result: A on B: +20 N on B's diagram. B on A: -20 N on A's diagram. Same size, opposite sign, different bodies.

3. **Ask whether either single body's force sum contains both partner arrows.**

Why valid: A force diagram for one body includes only forces acting on that body.

Result: third-law partners do not cancel inside one body's F_{net} unless another separate force happens to balance one.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

In space, a 70 kg astronaut pushes a 350 kg satellite with a force of 140 N. During the push, find the force on each body and each body's acceleration.

1. The interaction is the contact between hand and satellite: astronaut on satellite and satellite on astronaut.
2. Astronaut on satellite: 140 N, on the satellite's diagram. Satellite on astronaut: 140 N the opposite way, on the astronaut's diagram.
3. Satellite: $a = 140/350 = 0.4 \text{ m/s}^2$.
4. Astronaut: $a = 140/70 = 2 \text{ m/s}^2$, in the opposite direction. Neither diagram contains both forces, so they do not cancel.

Each body feels 140 N, in opposite directions: the satellite accelerates at 0.4 m/s^2 away from the astronaut, and the astronaut at 2 m/s^2 the other way.

A tempting wrong path

- Action and reaction cancel each other, so interacting bodies cannot accelerate.

Diagnose

- When a person pushes a cart, do the person's force on the cart and the cart's force on the person both appear on the cart's free-body diagram?

Repair

- No. Put each partner on the body it acts on; only then sum forces for one body.

Check it independently

- Multiply each body's mass by its acceleration: both products must equal the size of the one shared interaction force.
- The two accelerations must point opposite ways, with sizes in the inverse ratio of the masses.

Exit task

A swimmer pushes water backward. Identify the third-law partner force and state which body each force acts on.

Model answer

The swimmer pushes the water backward, and the water pushes the swimmer forward with an equal-magnitude opposite force. The two forces act on different bodies.

1. The interaction is swimmer-water contact.
2. Swimmer on water acts on the water.
3. Water on swimmer acts on the swimmer in the opposite direction.

Quantitative static and kinetic friction conditions

You need

- Can determine friction direction from relative slip tendency.
- Can write Newton-II equations for a chosen body.
- [Identify friction as a contact force that opposes relative sliding or the tendency to slide.](#)
- [Relate signed net external force to acceleration with \$F_{\text{net}} = m a\$.](#)

The key step

Static friction is a responsive contact force: solve for the friction required to prevent relative slip and compare it with the limiting value $\mu_s N$. Equality is a threshold condition, while the kinetic model is used only after sliding is established.

Find the actual normal force and the friction that rest would need, compare that need with $\mu_s N$, and use μN as the friction value only at impending slip or once sliding.

1. **Build the chosen-body FBD and determine the normal reaction N from the force balance or acceleration perpendicular to the contact; do not assume $N = mg$.**

Why valid: The normal reaction is the contact force needed by the actual perpendicular dynamics and can differ from mg when the surface is inclined or other forces have perpendicular components.

Result: perpendicular force equation $\rightarrow$ actual $N \geq 0$

2. **Assume no relative sliding provisionally and solve the tangential Newton-II equation for the static friction value f_{required} that would make that motion possible.**

Why valid: Static friction takes the value required by the no-slip state, provided the contact can supply it.

Result: no-slip dynamics $\rightarrow f_s = f_{\text{required}}$

3. **Compare $|f_{\text{required}}|$ with $\mu_s N$.**

Why valid: The static-contact model permits any required value only up to the limiting magnitude.

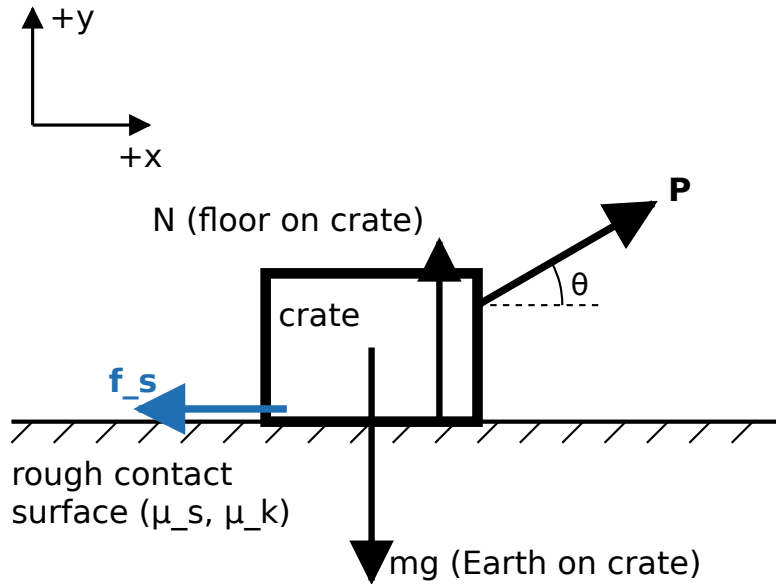
Result: $|f_{\text{required}}| \leq \mu_s N \rightarrow$ static state feasible; $|f_{\text{required}}| > \mu_s N \rightarrow$ static state impossible

4. **Use $|f_s| = \mu_s N$ only at impending slip; if relative sliding is already established, use $|f_k| = \mu_k N$ with direction opposing the relative sliding.**

Why valid: The limiting equality and the sliding model describe different contact states and are not unconditional identities.

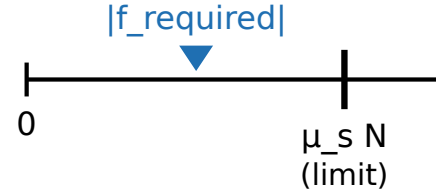
Result: impending slip $\rightarrow |f_s| = \mu_s N$; sliding $\rightarrow |f_k| = \mu_k N$

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



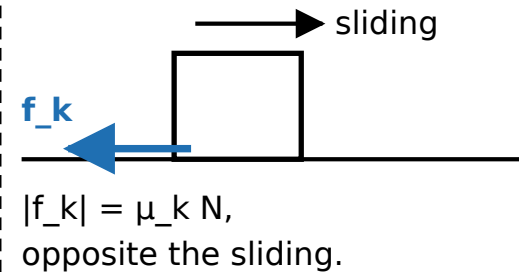
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Worked example

A 4 kg box rests on a rough horizontal floor with $\mu_s = 0.5$ and $\mu_k = 0.4$. Take $g = 10 \text{ m/s}^2$. A horizontal pull of 15 N is applied, and in a second trial a pull of 25 N. For each trial decide whether the box stays at rest, and find the friction force and the acceleration.

1. Vertical balance gives $N = mg = 4 \times 10 = 40 \text{ N}$, so static friction can supply at most $\mu_s N = 0.5 \times 40 = 20 \text{ N}$.
2. 15 N trial: rest would need friction of 15 N, and $15 \text{ N} \leq 20 \text{ N}$, so the box stays at rest with $f_s = 15 \text{ N}$ and $a = 0$.
3. 25 N trial: rest would need 25 N of friction, more than the 20 N available, so the box slides.
4. Once sliding, $f_k = \mu_k N = 0.4 \times 40 = 16 \text{ N}$ against the motion, so $a = (25 - 16)/4 = 2.25 \text{ m/s}^2$ in the direction of the pull.

At 15 N the box stays at rest with 15 N of static friction. At 25 N it slides, kinetic friction is 16 N, and the acceleration is 2.25 m/s^2 .

A tempting wrong path

- Friction always equals μN .
- The normal force is always mg .
- If both μ_s and μ_k are given, the coefficients themselves tell you which friction model to use.

Diagnose

- A small horizontal force is applied to a block that remains at rest. Must static friction already equal $\mu_s N$?
- If an extra force presses a block into a horizontal surface, can the normal reaction still be assumed equal to mg ?
- A problem gives both μ_s and μ_k but does not say whether the contact slips. Can you choose kinetic friction just because μ_k is supplied?

Repair

- No. First solve the force required for no slip. Static friction matches that requirement up to its limit; equality occurs only at impending slip.
- Determine N from the perpendicular free-body equation. $N = mg$ is only a special case.
- No. Determine the contact state first: test whether the no-slip friction demand fits within $\mu_s N$. Use the kinetic model only after sliding is established.

Check it independently

- Put the friction and acceleration back into the horizontal equation: the pull minus the friction must equal mass times acceleration in each case.
- Set the pull to zero: the method must return zero friction, not $\mu_s N$, and zero acceleration.

Exit task

A contact has known normal reaction N and coefficient μ_s . The no-slip force equation requires tangential friction f_{required} . State the test for whether static contact is possible, when equality with $\mu_s N$ is justified, and what changes once sliding is established.

Model answer

Static contact is possible when $|f_{\text{required}}| \leq \mu_s N$. Equality $|f_s| = \mu_s N$ is justified only at impending slip. After relative sliding is established, use the kinetic model $|f_k| = \mu_k N$ with direction opposite the relative sliding.

1. Static friction supplies the required no-slip value rather than an automatic maximum.
2. The limiting value $\mu_s N$ is a bound and becomes an equality only at the threshold of slipping.
3. The kinetic model belongs to a different contact state: established relative sliding.

Connected bodies and a justified common acceleration

You need

- Can draw one FBD per chosen body.
- Can apply signed Newton II to one chosen body.
- [Relate signed net external force to acceleration with \$F_{\text{net}} = m a\$.](#)

The key step

Bodies share an acceleration component only when a physical constraint keeps their relative separation fixed along that coordinate; the shared acceleration can link separate Newton-II equations, while internal partner forces cancel only if the bodies are deliberately treated as one system.

Name the physical link that fixes the separation, give both bodies one sign convention, and write a separate Newton-II equation for each body with the interaction force kept in it.

1. **Identify the connection/contact condition and state exactly which relative separation is fixed.**

Why valid: A common acceleration is a kinematic consequence of a constraint, not a default property of nearby bodies.

Result: fixed relative separation along chosen coordinate $\rightarrow$ common constrained acceleration component

2. **Choose one compatible sign convention for every body equation and assign the constrained acceleration component consistently.**

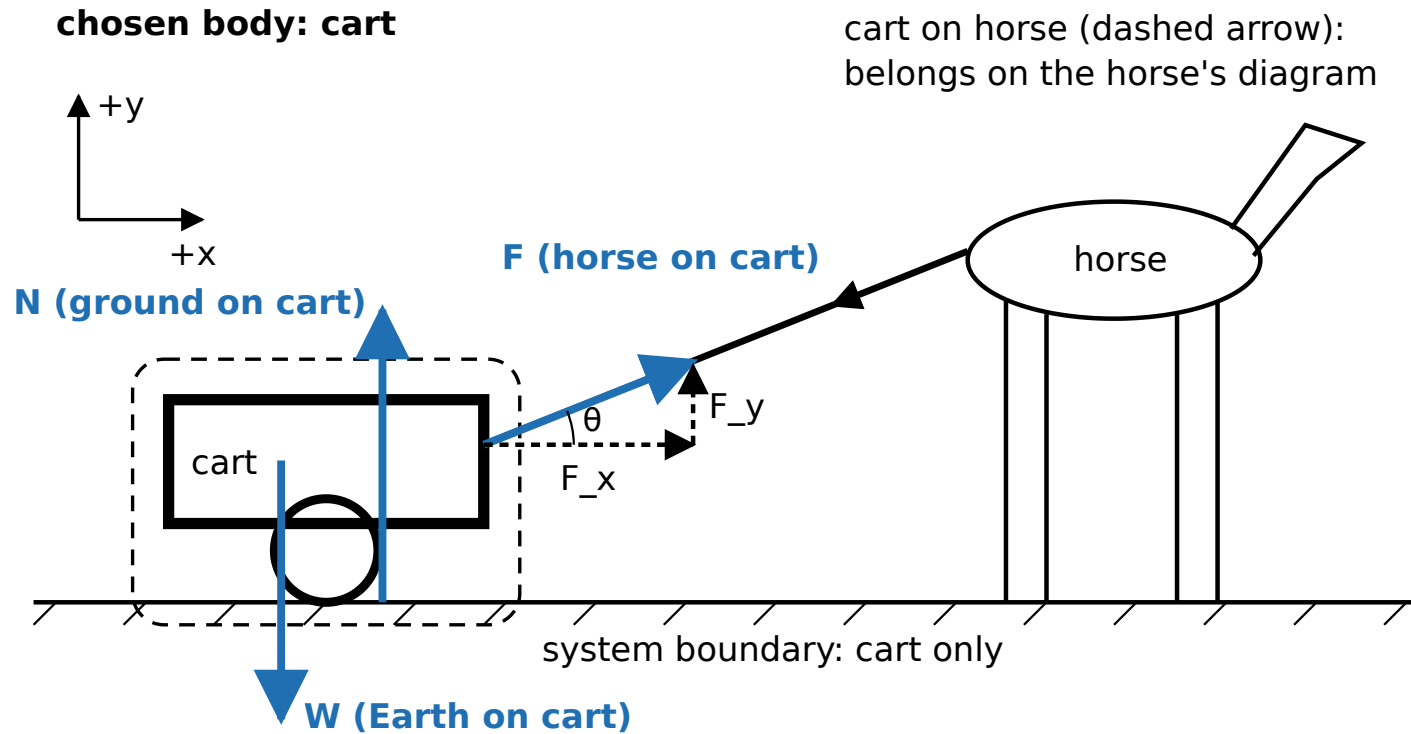
Why valid: The same physical acceleration cannot change sign merely because the equation changed bodies.

Result: same-sense coordinates $\rightarrow a_A = a_B = a$

3. **Write one Newton-II force sum for each body, keeping every inter-body force on the body it acts on.**

Why valid: The interaction forces are external to each individual body and therefore belong in its own equation.

Result: body A: $\Sigma F_A = m_A a$; body B: $\Sigma F_B = m_B a$



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

Blocks A (3 kg) and B (2 kg) touch on a smooth floor, A to the left of B. A 20 N force pushes A to the right into B, and a 5 N force pushes B to the left, back toward A. The blocks stay in contact. Write one Newton-II equation per block and solve for their acceleration and the contact force.

1. Maintained contact fixes the separation of A and B, so with rightward positive $a_A = a_B = a$.
2. For A: the 20 N push acts right and B pushes back on A with the contact force N , so $20 - N = 3a$.
3. For B: A pushes B right with N and the 5 N force acts left, so $N - 5 = 2a$.
4. Adding the equations removes N : $15 = 5a$, so $a = 3 \text{ m/s}^2$. Then $N = 5 + 2 \times 3 = 11 \text{ N}$.

The blocks accelerate together at 3 m/s^2 to the right, and the contact force between them is 11 N.

A tempting wrong path

- Any two touching or connected bodies automatically have the same acceleration.
- The contact force between two bodies can be cancelled from each individual FBD because it is an internal force of the pair.
- Bodies with the same acceleration must have the same net force.

Diagnose

- If two blocks are touching but one loses contact, what keeps their accelerations equal after separation?
- On body A alone, is the force from body B external to A?
- Two connected bodies of different masses share acceleration a . Must their individual net forces be equal?

Repair

- Name the active constraint first. Equal acceleration is justified only while the relative-separation condition remains enforced.
- Yes. Keep it in A's equation. It cancels only after A and B are deliberately combined into one system equation.
- No. The constraint can make their accelerations equal while Newton II gives different net forces $m_A a$ and $m_B a$ when the masses differ.

Check it independently

- Substitute the acceleration and the contact force into both body equations; each must balance on its own.
- The contact force must come out positive in the pushing sense: an ordinary contact can only push, so a negative value means the bodies have separated.

Add the body equations into one system equation only where the internal pair should drop out, then isolate the body that makes the requested interaction force external.

- 1. If treating A+B as one system, add the two body equations and cancel only the internal A-on-B/B-on-A pair in that system sum.**

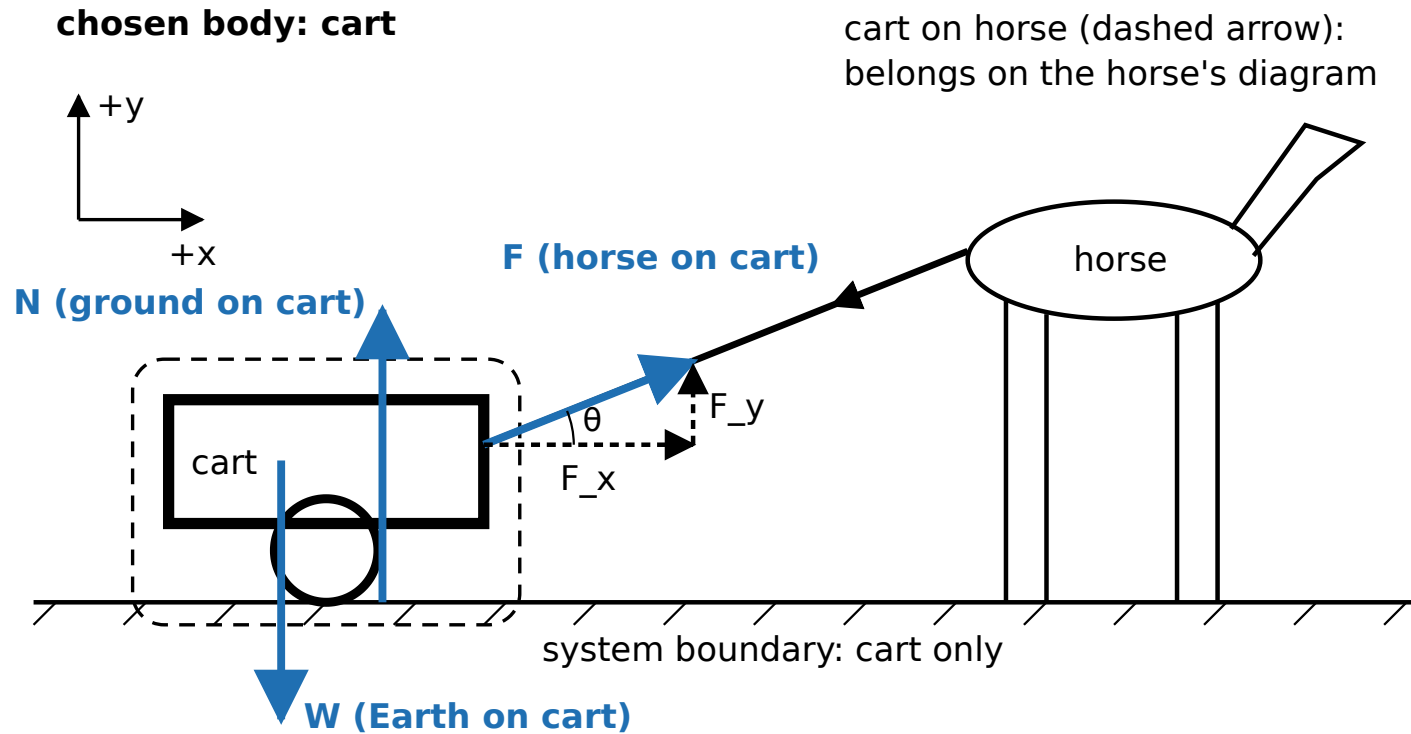
Why valid: Internal partner forces cancel in the combined system equation because both members are included, not because either force is absent from a body FBD.

Result: system: $\Sigma F_{\text{external}} = (m_A + m_B) a$; internal pair absent only after equations are combined

- 2. Match the chosen system boundary to the requested quantity: use the combined constrained system when internal forces are irrelevant to the shared acceleration, then isolate the body or subsystem that makes a requested interaction force external.**

Why valid: An internal interaction disappears from a whole-system force sum, so that boundary is efficient for acceleration but cannot by itself determine the internal force that was cancelled.

Result: target shared acceleration $\rightarrow$ whole constrained system; target internal/contact force $\rightarrow$ isolate a body or subsystem while carrying the same justified acceleration.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

A 2 kg bucket hangs from a light rope tied under a 1 kg bucket. A second rope lifts the top bucket with a 45 N upward force. Take $g = 10 \text{ m/s}^2$ and up as positive. Choose a system to find the acceleration, then a different system to find the tension in the rope between the buckets.

1. For the acceleration, take both buckets as one system: the connecting rope's pulls are internal and drop out.
2. External forces on the system: 45 N up and weight $(1 + 2) \times 10 = 30 \text{ N}$ down, so $45 - 30 = 3a$ and $a = 5 \text{ m/s}^2$.
3. For the connecting rope, isolate the lower bucket: that rope's pull is external to it.
4. Lower bucket: $T - 2 \times 10 = 2 \times 5$, so $T = 30 \text{ N}$.

The buckets accelerate upward at 5 m/s^2 , and the rope between them carries 30 N.

A tempting wrong path

- The contact force between two bodies can be cancelled from each individual FBD because it is an internal force of the pair.
- Bodies with the same acceleration must have the same net force.

Diagnose

- On body A alone, is the force from body B external to A?
- Two connected bodies of different masses share acceleration a . Must their individual net forces be equal?

Repair

- Yes. Keep it in A's equation. It cancels only after A and B are deliberately combined into one system equation.
- No. The constraint can make their accelerations equal while Newton II gives different net forces $m_A a$ and $m_B a$ when the masses differ.

Check it independently

- Check the body you did not use: its own forces must give its mass times the same acceleration.
- Reduce the driving force until the acceleration is zero: the internal force must then equal what simple balance of the isolated body gives.

Exit task

Two bodies are constrained to keep a fixed separation along one straight coordinate. State the acceleration relation, write the form of the two body equations, and explain when their interaction force can disappear from an equation.

Model answer

With same-sense coordinates the constraint gives $a_A = a_B = a$. Each body keeps the interaction force in its own Newton-II equation; the equal-and-opposite pair cancels only in a combined A+B system equation.

1. Fixed relative separation implies equal velocity and acceleration components along the constrained coordinate.
2. Each individual body experiences the partner force as an external force.
3. When both bodies are included in one system, the internal partner pair cancels in the summed equation.

Ideal-string tension belongs to each body-specific FBD

You need

- Can assign forces to the body they act on.
- Can apply Newton II to a chosen body.
- [Assign each force arrow to the body on which that force acts.](#)
- [Relate signed net external force to acceleration with \$F_{\text{net}} = m a\$.](#)

The key step

For one taut massless ideal string, with any redirection treated as frictionless and massless, the tension magnitude can be represented by one T throughout that same string; each body still receives its own tension force along the local string direction.

Declare the ideal string before writing a single T , then draw that tension on each attached body, pulling away from the body along its own segment.

1. **State the idealization before using one tension symbol: taut inextensible massless string, with any pulley/redirection frictionless and massless for this bounded model.**

Why valid: Uniform tension is a consequence of the ideal model and must not be silently transferred to a nonideal rope or pulley.

Result: ideal single string model declared -> one tension magnitude T is admissible throughout that string

2. For each attached body, draw the string's pull along the local string direction away from the body.

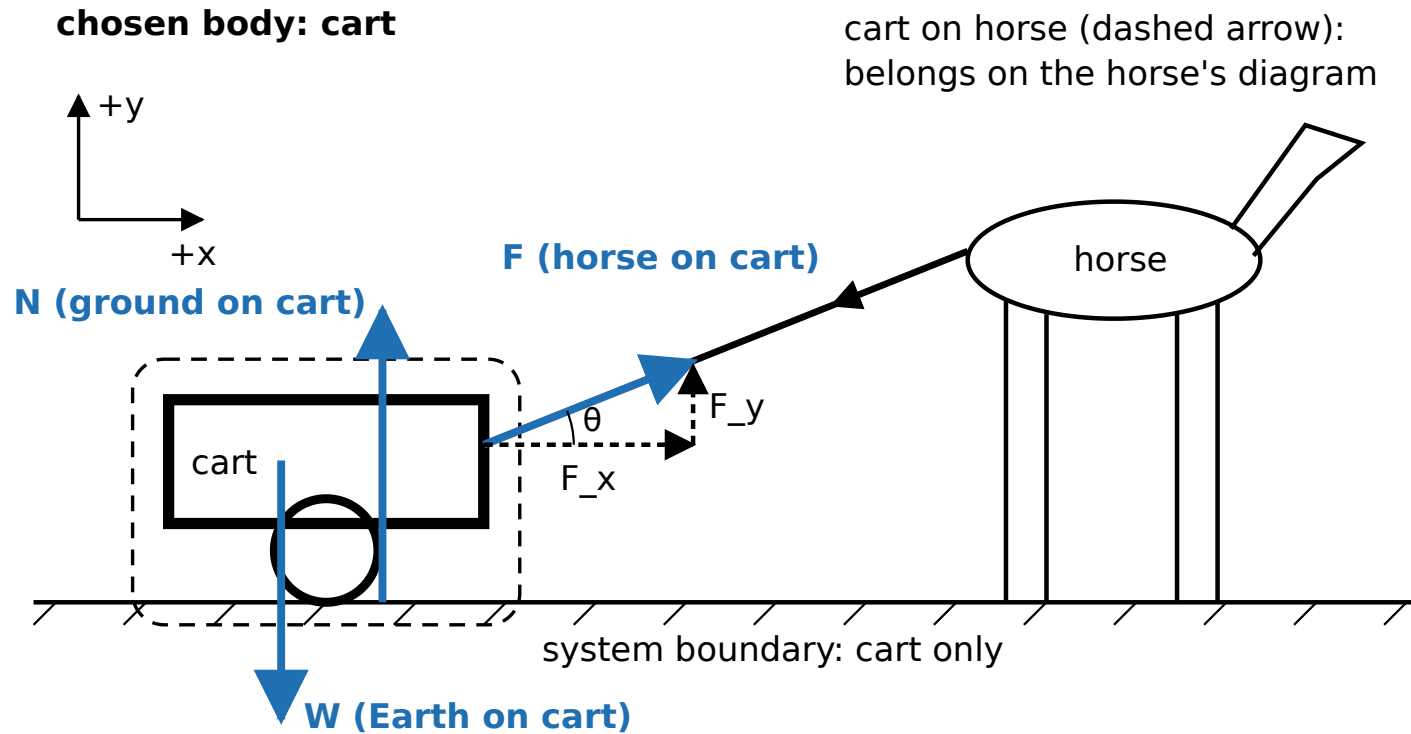
Why valid: Tension is a force exerted by the string on that body and belongs on that body's FBD.

Result: body A FBD contains string-on-A force of magnitude T ; body B FBD contains string-on-B force of magnitude T

3. Use the same symbol T for those magnitudes only while the declared ideal assumptions hold.

Why valid: A massless string segment cannot sustain a nonzero net force in this idealized model, and a frictionless massless redirection does not create a tension-magnitude jump.

Result: $T_A = T_B = T$ under the declared ideal-string/redirection conditions



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

A light, taut, inextensible string runs from a 2 kg cart on a smooth table, over a frictionless light pulley at the table's edge, down to a 1 kg hanging mass. Take $g = 10 \text{ m/s}^2$. State the model that allows one tension T , list the forces on each body, and find the acceleration and T .

1. The string is light, taut and inextensible and the pulley is light and frictionless, so one tension T acts along the whole string.
2. On the cart: T pulls it toward the pulley; its weight and the table's normal force balance vertically.
3. On the hanging mass: T pulls it up and its 10 N weight pulls it down; both bodies share one acceleration magnitude a .
4. Cart: $T = 2a$. Hanging mass, down positive: $10 - T = 1 \times a$. Adding gives $10 = 3a$, so $a = 10/3 \text{ m/s}^2$ and $T = 20/3 \text{ N}$.

One T is allowed by the ideal string and pulley. The acceleration is $10/3 \approx 3.33 \text{ m/s}^2$ and $T = 20/3 \approx 6.67 \text{ N}$.

A tempting wrong path

- Tension forces at the two ends of a string cancel because they are equal and opposite.

- Every rope over every pulley has the same tension on both sides.

Diagnose

- Do the two end tensions act on the same chosen body?
- What model assumptions justify using one symbol T throughout the rope?

Repair

- No. Put the tension force acting on each body in that body's own FBD. Equal magnitudes across the ideal string do not make them a cancelling pair on one body.
- State the ideal massless string and frictionless/massless redirection assumptions first. Without them, equal tension is not automatic.

Check it independently

- The hanging mass accelerates downward, so the tension on it must come out smaller than its weight; a tension above the weight would make it rise.
- Every term of each body equation must be a force in newtons: a mass times an acceleration, a weight, or the tension.

Check that nothing breaks the ideal model, then take the tension from a chosen body's signed Newton-II equation instead of setting it equal to a weight or an

applied force.

1. **Check whether the problem introduces rope mass, elasticity, pulley friction, or rotational inertia; if it does, stop using equal tension unless a different model justifies it.**

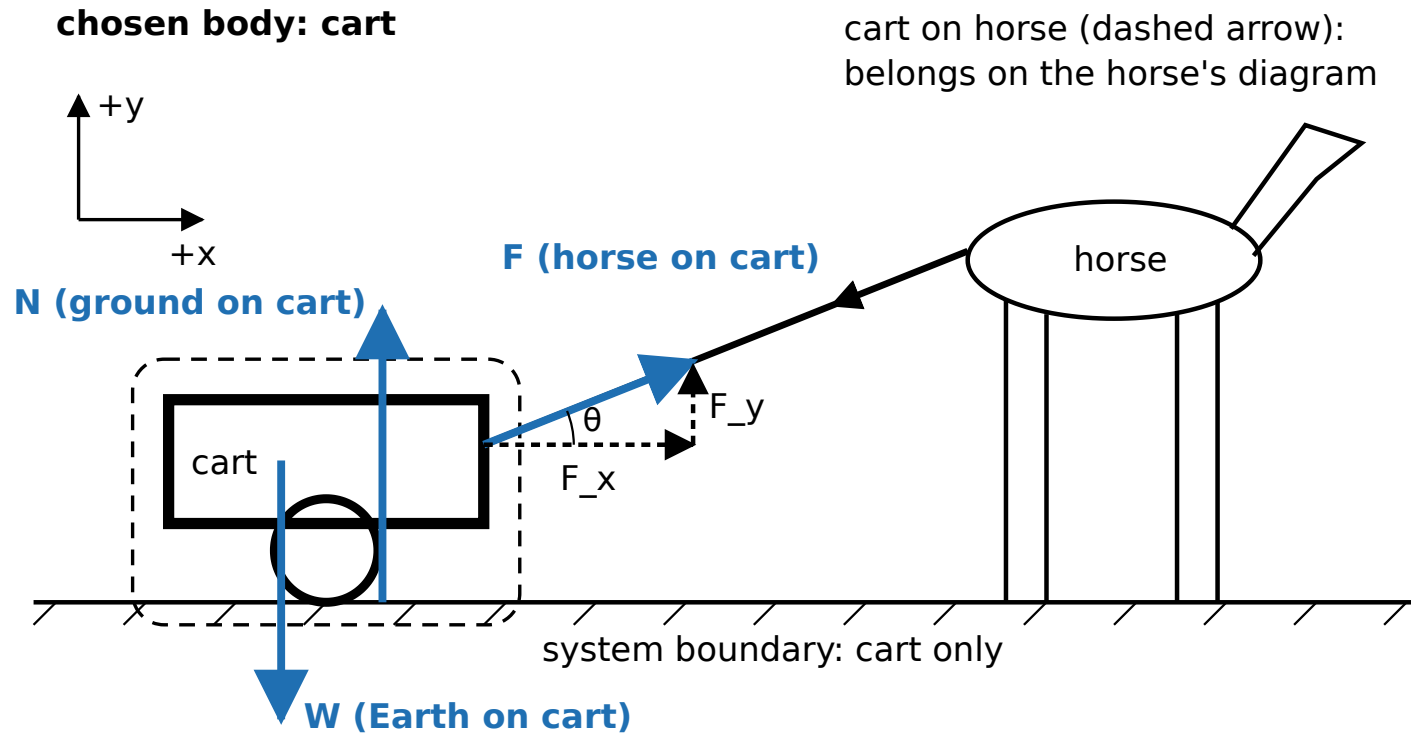
Why valid: Those changes alter the force transmission assumptions that supported one common T .

Result: nonideal feature present $\rightarrow$ equal-tension relation not licensed by this capability

2. **Determine the tension magnitude from a chosen body's signed Newton-II equation; do not replace T by mg or by an applied force merely because those forces appear in the same setup.**

Why valid: Uniform tension says the ideal string transmits one magnitude throughout itself; it does not say that magnitude equals some different force. For a hanging body, $T=mg$ only in the special zero-acceleration case.

Result: for $+y$ upward on a hanging mass: $T - mg = ma_y$; therefore $a_y=0 \rightarrow T=mg$, $a_y>0 \rightarrow T>mg$, and $a_y<0 \rightarrow T<mg$.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

A 2 kg lamp hangs from a light cord inside a lift. Take up as positive and $g = 10 \text{ m/s}^2$. Find the cord tension when the lift (a) moves up at steady speed, (b) accelerates upward at 2 m/s^2 , and (c) accelerates downward at 3 m/s^2 .

1. Only the cord and the lamp's weight act on the lamp, and nothing makes the cord non-ideal, so T comes from the lamp's own equation: $T - 20 = 2a$.
2. (a) Steady speed means $a = 0$, so $T = 20 \text{ N}$.
3. (b) $a = +2 \text{ m/s}^2$, so $T = 20 + 2 \times 2 = 24 \text{ N}$.
4. (c) $a = -3 \text{ m/s}^2$, so $T = 20 + 2 \times (-3) = 14 \text{ N}$.

The tension is 20 N, 24 N and 14 N in the three cases.

A tempting wrong path

- Every rope over every pulley has the same tension on both sides.
- The tension in a string holding a hanging mass is always equal to the mass's weight.

Diagnose

- What model assumptions justify using one symbol T throughout the rope?
- A hanging mass accelerates upward. Can T still equal mg while Newton II gives a nonzero upward net force?

Repair

- State the ideal massless string and frictionless/massless redirection assumptions first. Without them, equal tension is not automatic.
- Write the signed body equation. With upward positive, $T - mg = ma$, so upward acceleration requires $T > mg$; equality belongs only to zero vertical acceleration.

Check it independently

- Let the downward acceleration reach g : the equation must give zero tension, a slack cord.
- Upward acceleration must raise the tension above the weight and downward acceleration must lower it; a tension on the wrong side of mg shows a sign error.

Exit task

Two bodies are attached to the same taut ideal string, redirected only by a frictionless massless fixed pulley. State the tension relation and explain where the two tension forces appear.

Model answer

Under the declared ideal model, both string segments have the same tension magnitude T . Each body has its own string-on-body tension force on its own FBD; the forces are not cancelled across separate body diagrams.

1. The ideal assumptions license one common tension magnitude throughout the same string.
2. Each tension force acts on a different body and points along that body's local string segment.
3. Equal magnitude does not mean the forces cancel in an individual-body equation.

One fixed-pulley string gives one length constraint

You need

- Can identify a common-acceleration connection when separation is fixed.
- Can use an ideal single-string tension model when forces are also being solved.
- [Identify which bodies are constrained to share one acceleration component and write consistent body-specific or whole-system Newton-II equations.](#)
- [Use one declared ideal-string model to assign tension forces and, where justified, one common tension magnitude throughout that same string.](#)

The key step

For one taut inextensible string over one fixed ideal pulley, the two changing straight segment lengths add to a constant; with coordinates measured away from the pulley this gives $y_A + y_B = \text{constant}$ and therefore $a_A + a_B = 0$.

Measure both coordinates away from the fixed pulley, write the changing length as $y_A + y_B = \text{constant}$, and differentiate that one relation.

1. **Restrict the model to one taut inextensible string over one fixed ideal pulley and choose y_A and y_B positive away from the pulley along the two straight segments.**

Why valid: The coordinate declaration determines the signs in the length relation and blocks accidental import of movable-pulley rules.

Result: single fixed pulley; coordinates y_A, y_B measured away from pulley

2. **Write the variable part of the string length as $y_A + y_B$ and set its total equal to a constant.**

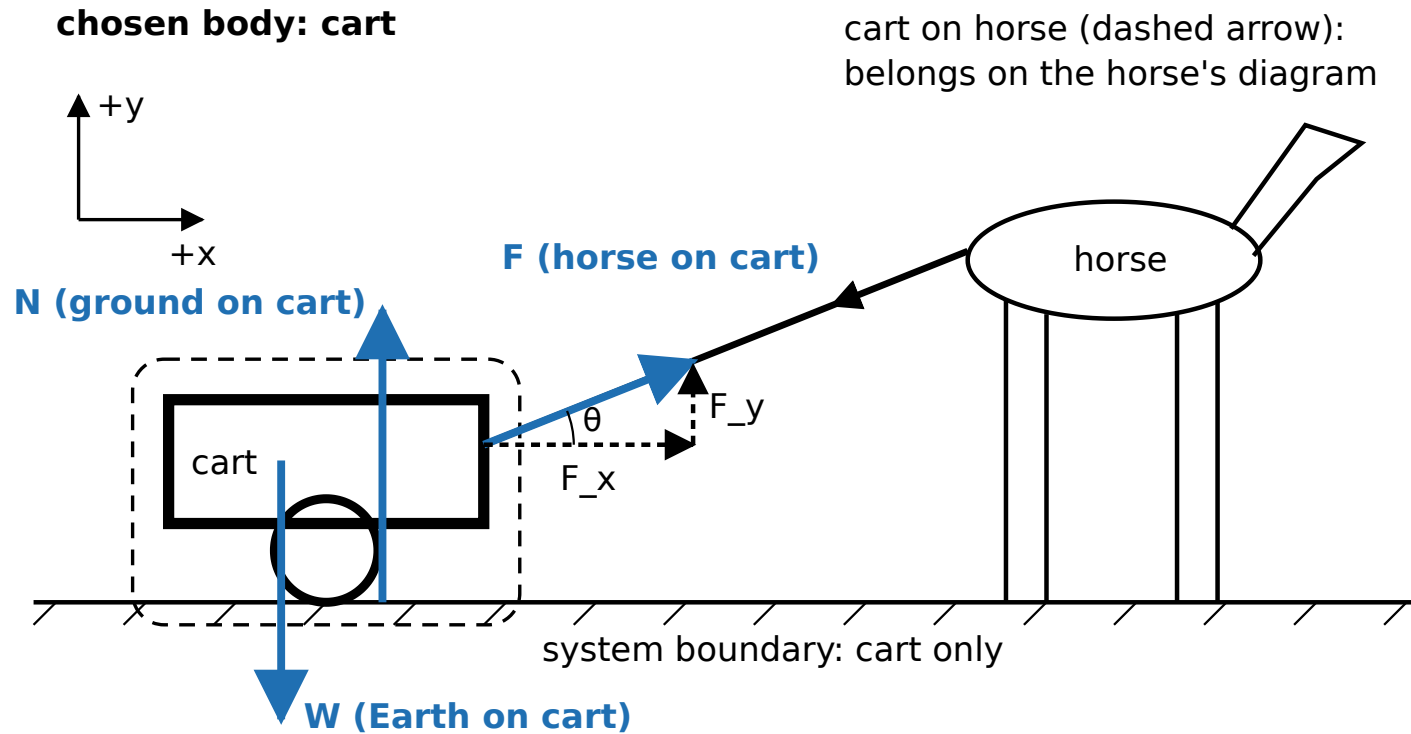
Why valid: An inextensible string has fixed total length; the wrapped portion around a fixed pulley is constant in this bounded geometry.

Result: $y_A + y_B = L_{\text{free}} = \text{constant}$

3. **Differentiate the fixed-length relation at the kinematic level needed by the problem.**

Why valid: A constant length has zero time derivative at each order.

Result: $v_A + v_B = 0$ and $a_A + a_B = 0$



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

One taut inextensible string passes over a fixed pulley and holds A and B. With y_A and y_B measured downward from the pulley along each segment, $y_A = 1.2$ m and $y_B = 0.8$ m. A then moves down by 0.3 m. Find B's new position, and B's velocity at an instant when A moves down at 0.5 m/s.

1. Both coordinates are measured away from the pulley, so the changing length is $y_A + y_B = 1.2 + 0.8 = 2.0$ m, fixed.
2. After A moves down 0.3 m, $y_A = 1.5$ m, so $y_B = 2.0 - 1.5 = 0.5$ m: B has risen 0.3 m.
3. Differentiating $y_A + y_B = 2.0$ m gives $v_A + v_B = 0$.
4. With $v_A = +0.5$ m/s, $v_B = -0.5$ m/s: B moves toward the pulley at 0.5 m/s.

B rises to $y_B = 0.5$ m; when A moves down at 0.5 m/s, $v_B = -0.5$ m/s, that is 0.5 m/s upward.

A tempting wrong path

- Every pulley problem has the same acceleration ratio.
- Bodies on opposite sides of a fixed pulley have the same signed acceleration.

Diagnose

- What part of the string length changes in this exact geometry, and is any pulley itself moving?
- If both coordinates are positive away from the pulley, can both segment lengths increase while total string length stays fixed?

Repair

- Write the string-length equation from the actual geometry. The equal-magnitude result here is only for one fixed pulley with two changing straight segments.
- No. Their signed changes are opposite: $a_A + a_B = 0$ in these coordinates.

Check it independently

- Add the new segment lengths: they must still give the original total.
- If both coordinates grew at once the string would have to stretch, so one must shrink while the other grows.

Read the signed result back as motion, and keep the length argument for the accelerations separate from the ideal-pulley argument for equal tension.

1. **Translate the signed result back into motion: if A accelerates farther from the pulley, B accelerates toward it with equal magnitude.**

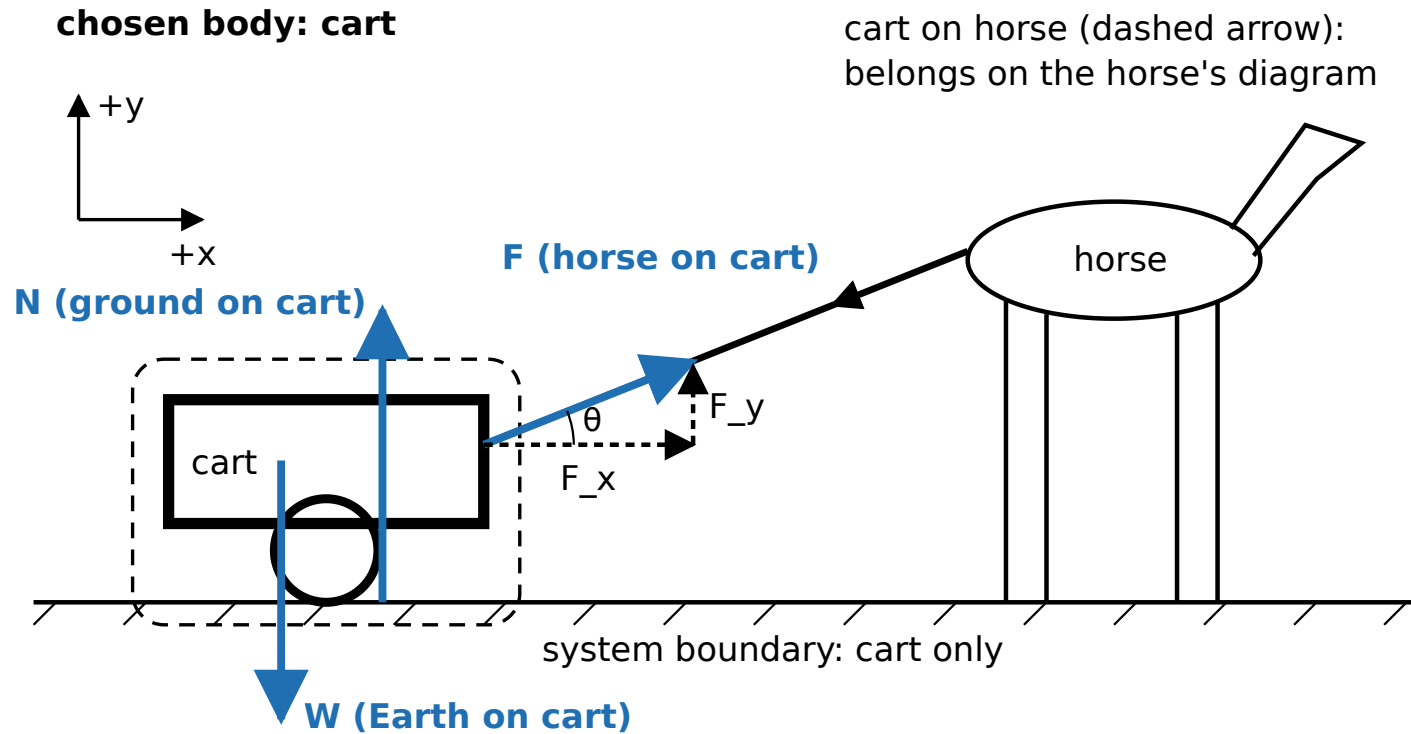
Why valid: The two segment changes must cancel to preserve total string length.

Result: $|a_A| = |a_B|$ with opposite signs in the declared away-from-pulley coordinates

2. **Keep the kinematic string-length claim separate from the force-transmission claim: derive the acceleration relation from inextensibility and fixed geometry, and use equal tension only when the independent ideal-string/redirection assumptions license it.**

Why valid: The acceleration relation comes from constant string length, whereas tension equality comes from the force model. One statement is not the proof of the other.

Result: fixed-length geometry $\rightarrow a_A + a_B = 0$ in the declared coordinates; ideal force-transmission model $\rightarrow T_A = T_B$. Treat these as separate justified statements.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Worked example

Masses A (3 kg) and B (1 kg) hang from one light inextensible string over a fixed, frictionless, light pulley, with y_A and y_B measured downward from the pulley. Take $g = 10 \text{ m/s}^2$. Using $a_A + a_B = 0$ and one tension T , find a_A , a_B and T , and name the assumption each of those two relations rests on.

1. The fixed total length gives $a_A + a_B = 0$ in these coordinates; this uses geometry only.
2. One tension T is licensed separately, because the string is light and the pulley is light and frictionless.
3. Down positive for each mass: A: $30 - T = 3a_A$. B: $10 - T = 1 \times a_B = -a_A$.
4. Subtracting B's equation from A's: $20 = 4a_A$, so $a_A = 5 \text{ m/s}^2$, $a_B = -5 \text{ m/s}^2$, and $T = 30 - 3 \times 5 = 15 \text{ N}$.

$a_A = +5 \text{ m/s}^2$ (down), $a_B = -5 \text{ m/s}^2$ (up) and $T = 15 \text{ N}$. The acceleration relation rests on the fixed string length; the single T rests on the ideal pulley and string.

A tempting wrong path

- Bodies on opposite sides of a fixed pulley have the same signed acceleration.
- The two end accelerations are equal and opposite because the string tension is equal on both sides.

Diagnose

- If both coordinates are positive away from the pulley, can both segment lengths increase while total string length stays fixed?
- If equal tension were no longer guaranteed but the same fixed, taut, inextensible string still could not change length, which relation would you re-check from geometry?

Repair

- No. Their signed changes are opposite: $a_A + a_B = 0$ in these coordinates.
- Derive the acceleration relation from the fixed-length equation. Equal tension is a separate force-model statement and is not the cause of the kinematic constraint.

Check it independently

- Treat both masses as one system driven by the difference of their weights: the acceleration must match the one from the two body equations.
- Make the two masses equal: both accelerations must be zero and the tension must equal the common weight.

Exit task

One taut inextensible string passes over one fixed ideal pulley, with y_A and y_B measured away from the pulley along the two hanging segments. Write the length relation and its acceleration consequence.

Model answer

The variable string length is $y_A + y_B = \text{constant}$, so $a_A + a_B = 0$. The two acceleration magnitudes are equal and their signs are opposite in the declared coordinates.

1. The fixed pulley contributes no changing geometric length in this bounded model.
2. Because the string is inextensible, the sum of the two straight segment lengths is constant.
3. Two time derivatives give the acceleration relation.